Lesson notes · for you

Integer programs and branch and bound

Optimization · Level 4 · Lesson 2

Time	Materials
25 min	Your phone, or a computer with Colab; Paper for drawing a branch-and-bound tree

Learning goals

- Explain why rounding an LP can fail, and why the relaxation is a bound.
- Follow branch and bound: branch, bound, prune, and the gap.
- Model whole units and a fixed cost in SCIP.

Activity: The bus tree by hand

1. Root (5 min): The cheapest cost per seat is the bus (550 ÷ 30 ≈ 18.3 against 300 ÷ 14 ≈ 21.4), so the relaxation uses buses only: 64 ÷ 30 = 2.13 buses, 1,173.33 coins. Write it at the top of a tree.
2. Branch (8 min): Draw two children: “at most 2 buses” and “at least 3 buses”. In the first, 2 buses seat 60, and the relaxation adds 4 ÷ 14 ≈ 0.29 of a minibus: about 1,186. In the second, 3 buses cost 1,650.
3. Prune (7 min): In the first child the minibus is fractional, so branch on it. “No minibus” is impossible (2 buses seat only 60). “At least 1 minibus” gives 1 minibus and 1.67 buses (about 1,217), so branch on buses: “at least 2 buses” gives 2 buses and 1 minibus, 1,400, a whole plan. The 3-bus branch can’t beat it (1,650), so cross it out without exploring it. The “at most 1 bus” branch (about 1,279) could still win, so pairs who finish early explore it: it holds only worse plans, 1 bus and 3 minibuses (1,450) or 5 minibuses (1,500).
4. Fee (5 min): Add a 300-coin booking fee for any minibus. Recompute: 2 buses + 1 minibus = 1,700, so 3 buses (1,650) win. At what fee does the answer flip? (250.)

Work in pairs: one draws the tree, the other checks each relaxation. Then run the same model in the lesson’s code step and compare.

Discussion

- Why can rounding an LP answer break a constraint?
- Why does a smaller big-M make branch and bound faster?
- When is a 3% gap good enough, and who decides?

Check yourself

- A relaxation costs 900 and your best plan 1,000. Write the gap, and what it promises.

Try it this week

Model the field trip in SCIP with the 300-coin booking fee (1,650). Then change the fee and find where the answer flips from three buses back to two buses and a minibus. Explain the flip point in one sentence.

Worksheet

Integer programs and branch and bound

Optimization · Level 4 · Lesson 2

Name _____ Class _____ Date _____

1 **64 people; a minibus seats 14 for 300 coins and a bus 30 for 550. Without fees the best plan is 2 buses and 1 minibus (1,400). The minibus company now charges a 300-coin booking fee if you hire any minibus. What is the best plan now?**

◯ Still 2 buses and 1 minibus (1,400) ◯ Three buses, 1,650: with the fee the old plan costs 1,700

◯ Hire five minibuses instead

2 **Put the lines of a branch-and-bound search for a knapsack in order. bound(i, room) is an optimistic estimate of what loads i onwards could still add. Explore taking a load before skipping it.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `w, v = items[i]`

☐ `if w <= room:`

☐ `if i == n or value + bound(i, room) <= best:`

☐ `best = min(best, value)`

☐ `def search(i, room, value):`

☐ `search(i + 1, room, value)`

☐ `search(i + 1, room + w, value - v)`

☐ `return`

☐ `global best`

☐ `best = max(best, value)`

☐ `search(i + 1, room - w, value + v)`

3 **In a minimisation problem, the LP relaxation's cost can never be higher than the cost of the best whole-number plan.**

◯ True ◯ False

4 **The minibus company now charges a 300-coin booking fee if you hire any minibus. Add a yes/no variable book, a limit so minibuses need a booking (at most 10 when booked), and the fee in the cost. What does the trip cost now?**

```
from ortools.linear_solver import pywraplp
s = pywraplp.Solver.CreateSolver("SCIP")
small = s.IntVar(0, 10, "minibuses")
big = s.IntVar(0, 10, "buses")
s.Add(14 * small + 30 * big >= 64)
s.Minimize(300 * small + 550 * big)
s.Solve()
print(int(s.Objective().Value()))
```

Write your program:

5 **Which of these decisions need whole-number or yes/no variables? Choose all that apply.**

- ☐ Litres of water for the garden each week
- ☐ Whether to open a new tutoring centre
- ☐ Kilograms of rice to buy for the canteen
- ☐ Which lecturer invigilates which exam
- ☐ How many buses to hire

✎ **Where have you seen a decision rounded when it should have been solved with whole numbers? What went wrong, or might have?**

Check your answers

Integer programs and branch and bound

Optimization · Level 4 · Lesson 2

1 **Three buses, 1,650: with the fee the old plan costs 1,700**

The fee is a fixed cost, modelled with a yes/no variable.

2 **Top to bottom: 6, 7, 4, ✗, 1, 9, ✗, 5, 2, 3, 8**

```
def search(i, room, value):
    global best
    best = max(best, value)
    if i == n or value + bound(i, room) <= best:
        return
    w, v = items[i]
    if w <= room:
        search(i + 1, room - w, value + v)
    search(i + 1, room, value)
```

Record the value reached so far, then prune: stop at the end of the list, or when even the optimistic bound can't beat the best.

3 **True**

Every whole-number plan is also allowed in the relaxation, so the relaxation's minimum is at most the integer minimum: 1,173.33 against 1,400 for the trip.

4

```
from ortools.linear_solver import pywraplp
s = pywraplp.Solver.CreateSolver("SCIP")
small = s.IntVar(0, 10, "minibuses")
big = s.IntVar(0, 10, "buses")
book = s.BoolVar("book")
s.Add(14 * small + 30 * big >= 64)
s.Add(small <= 10 * book)
s.Minimize(300 * small + 550 * big + 300 * book)
s.Solve()
print(int(s.Objective().Value()))
```

The big-M limit small <= 10 * book lets minibuses be hired only when book = 1, and 300 * book charges the fee exactly then.

5 **How many buses to hire; Whether to open a new tutoring centre; Which lecturer invigilates which exam**

Buses come whole, opening a centre is yes or no, and assigning a lecturer to an exam is a yes/no choice for each pair.